

Explaining Neural Networks in Simple Language

Neural networks are like a person learning from experience.

When we experience something new, we slowly begin to understand it. As we continue learning and practicing, we become more confident and are able to explain it to others.

Think about a child learning to recognize a dog. At first, the child cannot tell the difference between a dog and a cat. After seeing many pictures and being told, "This is a dog" and "This is a cat," the child starts noticing patterns such as the ears, fur, tail, and size. After enough practice, the child can recognize a dog it has never seen before because it has learned those patterns.

A neural network works in a similar way. Instead of learning from personal experience, it learns from data. The computer studies many examples, recognizes patterns, and gradually becomes better at making predictions or decisions.

A neural network can be explained in three simple steps:

- **Input:** The computer receives information such as a photo, voice recording, text, or numbers.
- **Learning:** The computer studies many examples, finds patterns, and improves whenever it makes mistakes.
- **Output:** The computer gives a prediction or decision, such as "This is a dog" or "This email is spam," based on what it has learned.

We use neural networks every day without even realizing it. For example, Face Unlock on our phones recognizes our face after learning our facial features during setup. Similarly, Netflix and YouTube recommend movies and videos by learning patterns from what we have watched before.

They are called neural networks because they are inspired by the human brain. Our brain contains billions of nerve cells called neurons that work together to process information. A computer's neural network is much simpler, but it also uses connected artificial neurons that learn patterns from data. The more high-quality examples a neural network learns from, the better it becomes at recognizing patterns and making accurate predictions.